

of my involvement with derivatives and I had trouble getting through to them some basic points about financial markets (they believed in their models a little too much). There was all along lurking in my mind the idea that these researchers had missed a point, but I did not quite know what it was. It was not what they knew, it was how they knew it that was the subject of my annoyance.

POPPER'S ANSWER

Popper came up with a major answer to the problem of induction (to me he came up with *the* answer). No man has influenced the way scientists do science more than Sir Karl – in spite of the fact that many of his fellow professional philosophers find him quite naïve (to his credit, in my opinion). Popper's idea is that science is not to be taken as seriously as it sounds (Popper when meeting Einstein did not take him as the demigod he thought he was). There are only two types of theories:

1. Theories that are known to be wrong, as they were tested and adequately rejected (he calls them *falsified*).
2. Theories that have not yet been known to be wrong, not *falsified* yet, but are exposed to be proved wrong.

Why is a theory never *right*? Because we will never know if all the swans are white (Popper borrowed the Kantian idea of the flaws in our mechanisms of perception). The testing mechanism may be faulty. However, the statement that there is a black swan is possible to make. A theory cannot be *verified*. To paraphrase baseball coach Yogi Berra again, *past data has a lot of good in it, but it is the bad side that is bad*. It can only be provisionally accepted. A theory that falls outside of these two categories is not a theory. A theory that does not present a set of conditions under which it would be considered wrong would be termed charlatanism – they would be impossible to reject otherwise. Why? Because the astrologist can always find a reason to fit the past event, by saying that *Mars was probably in line but not too much so* (likewise to me a trader who does not have a point that would make him change his mind is not a trader). Indeed the difference between Newtonian physics, which was falsified by Einstein's relativity, and astrology lies in the

following irony. Newtonian physics is scientific because it allowed us to falsify it, as we know that it is wrong, while astrology is not because it does not offer conditions under which we could reject it. Astrology cannot be disproved, owing to the auxiliary hypotheses that come into play. Such point lies at the basis of the demarcation between science and nonsense (called "the problem of demarcation").

More practically to me, Popper had many problems with statistics and statisticians. He refused to blindly accept the notion that knowledge can always increase with incremental information – which is the foundation of statistical inference. It may in some instances, but we do not know which ones. Many insightful people, such as John Maynard Keynes, independently reached the same conclusions. Sir Karl's detractors believe that favorably repeating the same experiment again and again should lead to an increased comfort with the notion that "it works". I came to understand Popper's position better once I saw the first rare event ravaging a trading room. Sir Karl feared that some type of knowledge did not increase with information – but which type we could not ascertain. The reason I feel that he is important for us traders is because to him the matter of knowledge and discovery is not so much in dealing with what we know, as in dealing with what we do not know. His famous quote:

These are men with bold ideas, but highly critical of their own ideas; they try to find whether their ideas are right by trying first to find whether they are not perhaps wrong. They work with bold conjectures and severe attempts at refuting their own conjectures.

"These" are scientists. But they could be anything.

Putting the master in context, Popper was rebelling against the growth of science. Popper intellectually came to the world with the dramatic shifts in philosophy as attempts were made to shift it from the verbal and rhetorical to the scientific and rigorous, as we saw with the presentation of the Vienna Circle in Chapter 4. These people were sometimes called the logical positivists, after the movement called *positivism* pioneered in France in the nineteenth century by Auguste Comte, where positivism meant scientification of things (literally everything under the sun). It was the equivalent of bringing the industrial revolution into the soft sciences.